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# CLINTON ENWEREM

Academic CV

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## RESEARCH INTERESTS

Robot Learning, Dexterous Robotic Grasping & Manipulation, Multi-Modal Uncertainty Estimation.

## EDUCATION

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| 2021–Fall 2026 | <b>Ph.D., Electrical &amp; Computer Engineering</b>   <i>University of Maryland, College Park</i> <span style="float: right;">MD, USA</span><br>Advisors: Professor <a href="#">John S. Baras</a> and Professor <a href="#">Calin Belta</a> . <b>Admitted to Candidacy:</b> June 2025.<br><i>Relevant Coursework:</i> Decision-Making for Robotics, Network Control Systems, Decision Making Under Uncertainty, Random Processes, Advanced Digital Signal Processing, Nonlinear Control Systems, Optimal Control, System Theory, Convex Optimization, Formal Methods for Control & Dynamical Systems, Multimodal Foundational Models. |
| 2014–2018      | <b>B.Eng., Electrical Engineering</b>   <i>University of Nigeria, Nsukka</i> <span style="float: right;">Enugu, Nigeria</span><br><i>Graduated with Highest Honors. Emphasis:</i> Control Theory. <i>Completed coursework in engineering numerical analysis, control systems, electrical machines and drives, computer architecture, electronics, and communication systems.</i>                                                                                                                                                                                                                                                      |

## PUBLICATIONS

### Preprints / Articles In Review

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| 2026 | [P1] <b>Clinton Enwerem</b> , John S. Baras, and Calin Belta. EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows, June 2026. <a href="#">arXiv Link</a> . |
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### Refereed Conference Publications

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| 2026 | [C6] <b>Clinton Enwerem</b> , John S. Baras, and Calin Belta. Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty, December 2026. <i>To appear in the Proceedings of the 65th IEEE Conference on Decision and Control (CDC)</i> . <a href="#">arXiv Link</a> .                                                                    |
| 2026 | [C5] <b>Clinton Enwerem</b> , Shreya Kalyanaraman, John S. Baras, and Calin Belta. Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty, October 2026. <i>To appear in the Proceedings of the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> . <a href="#">arXiv Link</a> . |
| 2025 | [C4] <b>Clinton Enwerem</b> , Aniruddh G. Puranic, John S. Baras, and Calin Belta. Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression. <i>2025 IEEE 64th Conference on Decision and Control (CDC)</i> , pages 4890–4895, December 2025. <a href="#">arXiv Link</a> .                                |
| 2024 | [C3] <b>Clinton Enwerem</b> , Erfaun Noorani, John S. Baras, and Brian M. Sadler. Robust Stochastic Shortest-Path Planning via Risk-Sensitive Incremental Sampling. <i>2024 IEEE 63rd Conference on Decision and Control (CDC)</i> , pages 1087–1094, December 2024. <a href="#">arXiv Link</a> .                                                                  |
| 2024 | [C2] <b>Clinton Enwerem</b> and John S. Baras. Safe Collective Control under Noisy Inputs and Competing Constraints via Non-Smooth Barrier Functions. <i>2024 European Control Conference (ECC)</i> , pages 3762–3768, June 2024.                                                                                                                                  |
| 2023 | [C1] <b>Clinton Enwerem</b> and John S. Baras. Consensus-Based Leader-Follower Formation Tracking for Control-Affine Nonlinear Multiagent Systems. <i>9th International Conference on Control, Decision and Information Technologies (CoDIT)</i> , July 2023.                                                                                                      |

### Refereed Journal Articles

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| 2024 | [J1] <b>Clinton Enwerem</b> and John S. Baras. Formation Tracking for a Class of Uncertain Multiagent Systems: A Distributed Kalman Filtering Approach. <i>IEEE Control Systems Letters</i> , volume 8, 2024. |
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## RESEARCH EXPERIENCE

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| 8/2021–Present<br>Research Topics<br>Duties          | <b>Graduate Research Assistant</b>   <i>Institute for Systems Research, University of Maryland Learning-Enabled Methods for Robust Dexterous Robotic Grasping &amp; Manipulation</i> <ul style="list-style-type: none"><li>Design learning-enabled grasping and manipulation algorithms for robotic arms with anthropomorphic end-effectors, spanning perception, belief estimation, planning, and control.</li><li>Develop and maintain performant and research-grade software in Python and C++ for grasp planning, stochastic motion planning, and closed-loop manipulation control.</li><li>Benchmark algorithms at scale in MuJoCo and Drake, and validate learned policies in OpenAI Gym and Safety Gymnasium across contact-rich manipulation tasks.</li><li>Disseminate findings through peer-reviewed conference and journal papers, with work accepted at CDC and IROS, and submitted to RAL.</li></ul> | College Park, MD |
| 9/2018–3/2021<br>Research Topics<br>Accomplishments  | <b>Research Assistant</b>   <i>Electrical Engineering Department, University of Nigeria Robust Control, Observer-Based Compensator Design, Feedback Linearization</i> <ul style="list-style-type: none"><li>Designed and implemented robust nonlinear control, active disturbance rejection, feedback linearization, and observer-based compensation for hardware motor control, released as open-source <a href="#">software</a>.</li><li>Co-authored the resulting peer-reviewed <a href="#">journal paper</a>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                       | Enugu, Nigeria   |
| 8/2017–10/2017<br>Research Topics<br>Accomplishments | <b>Undergraduate Research Assistant</b>   <i>EE Department, University of Nigeria Feedback Control, Time-Delayed Systems, System Identification</i> <ul style="list-style-type: none"><li>Designed feedback controllers for first-order-plus-dead-time processes with dead-time compensation and closed-loop system identification; <a href="#">implementation and write-up</a> available online.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Enugu, Nigeria   |

## PROFESSIONAL EXPERIENCE

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| Jun.–Aug. 2023<br>Supervisor<br>Accomplishments | <b>Summer Research Assistant</b>   <i>Institute for Systems Research, University of Maryland Professor John S. Baras</i> <ul style="list-style-type: none"><li>Formulated a multi-agent safety-critical control problem as a chance-constrained mathematical program.</li><li>Proposed a novel solution based on Boolean-composed control barrier certificates.</li><li>Wrote software to validate the approach and prepared a research paper to summarize results.</li></ul>                                                      | College Park, MD |
| 3/2020–2/2021<br>Accomplishments                | <b>Robotics Engineer</b>   <i>Kognitive Robotics</i><br><i>Kognitive Robotics is a robotics startup developing turnkey robotics education platforms.</i> <ul style="list-style-type: none"><li>Brought up a mobile robot end to end: designed its power and sensor-actuator interface electronics (Altium schematic capture and PCB layout) and integrated the sensing and actuation stack.</li><li>Developed the robot's ROS-based software stack and validated hardware-software integration on the physical platform.</li></ul> | Lagos, Nigeria   |

## TEACHING EXPERIENCE

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| 9/2025–12/2025<br>Course<br>Accomplishments | <b>Graduate Teaching Assistant</b>   <i>ECE Department, University of Maryland ENEE467: Robotics Project Laboratory</i> (with Prof. Calin Belta) <ul style="list-style-type: none"><li>Mentored 50+ undergraduate students in kinematics, 3D perception, motion planning, and ROS 2 control.</li><li>Built reproducible Docker-based labs and testing pipelines.</li><li>Led weekly lab sessions and provided technical feedback on documentation and debugging.</li></ul> | College Park, MD |
| 2/2025–5/2025<br>Course<br>Accomplishments  | <b>Graduate Teaching Fellow</b>   <i>ECE Department, University of Maryland ENEE661: Nonlinear Control Systems</i> (with Prof. John S. Baras) <ul style="list-style-type: none"><li>Led problem-solving sessions on nonlinear analysis and control, covering phase-plane and bifurcation analysis, and stability concepts including Lyapunov, input-to-state, <math>L_2</math>, and <math>L_\infty</math> stability.</li></ul>                                             | College Park, MD |

## TECHNICAL SKILLS

**Robotics:** ROS/ROS2, MuJoCo, Drake, RViz, MoveIt2; **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash,  $\text{\LaTeX}$ ; **Optimization:** Gurobi, Pyomo, Mosek, CVXPY; **RL Sandboxes:** OpenAI Gym, Safety Gymnasium, PyBullet; **Dev Tools:** git, GitHub, GitLab, Docker; **Robots & EoATs:** UR3e/UR5, xArm7, LEAP Hand, RealHand L6.

## HONORS & AWARDS

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| 2024      | IEEE CSS Student Travel and Workshop Support: Conference travel award to attend CDC'24.     |
| 2022      | '22-'23 Microsoft Diversity in Robotics & Autonomy PhD Fellowship, MRC & Microsoft.         |
| 2022      | ROSCon Diversity Scholarship: Travel grant to attend ROSCon 2022 in Kyoto, Japan.           |
| 2021      | Finalist, Engineers' League, Pan-African Robotics Competition, Rwanda.                      |
| 2021      | CIT Dean's Fellowship, Carnegie Mellon University, Africa Campus, Kigali, Rwanda.           |
| 2021      | Dean's Fellowship, University of Maryland, College Park, MD, United States.                 |
| 2020      | Scholar, Stanford Exposure to Research & Graduate Education, Stanford University, CA, USA.  |
| 2020      | EducationUSA Opportunity Funds Program Scholarship, U.S. Consulate General, Lagos, Nigeria. |
| 2020      | Sole Recipient (Nationwide), Door Foundation Leadlight Scholarship, RAIN.                   |
| 2015-2018 | MTN Foundation Scholarship: Nationwide merit-based undergraduate scholarship.               |

## PROFESSIONAL TRAINING & DEVELOPMENT

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| 1/2020-2/2021     | <b>Certificate in Robot Development &amp; Automation   RAIN</b>                                                                                                                                                                                                                                                                                                                            |
| S'22/'23/'25/F'25 | Completed graduate-level coursework and projects in robotics, control theory, ML, and IoT.<br><b>Open Courseware:</b> Bayesian Statistics ( <i>UCSD</i> , <i>Coursera</i> ), Autonomous Navigation for Flying Robots ( <i>TUM</i> , <i>edX</i> ), Principles of Robot Autonomy I & II ( <i>Stanford</i> ), Planning & Learning in Robotics ( <i>UCSD</i> ), Robot Learning ( <i>USC</i> ). |

## PROFESSIONAL SERVICE, OUTREACH, & MENTORING

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| 2023-2026    | <i>Peer Reviewer:</i> Heliyon, MED'23, ECC'24, ACC'25, L4DC'25, CDC'25, CDC'26, Elsevier (Robotics & Autonomous Systems).                                                                  |
| 2021-Present | <i>Member,</i> Black in Robotics (BiR): the first organization of its kind, advancing Black representation across academic and industry robotics.                                          |
| 2021-2022    | <i>Mentor,</i> EducationUSA & iScholar Initiative: Guided STEM graduates from underrepresented backgrounds to secure fully-funded graduate admissions at Stanford, Purdue, and Penn State. |